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ME 639 - Introduction to Robotics  
HW-1

P-1

Let:  $a = [a_1 \ a_2 \ a_3]^T$

$b = [b_1 \ b_2 \ b_3]^T$

$c = [c_1 \ c_2 \ c_3]^T$

a)  $a \cdot (b \times c) = (a \times b) \cdot c$

LHS:

$a \cdot (b \times c)$

$$b \times c = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= (c_3 b_2 - c_2 b_3) \hat{i} - (b_1 c_3 - c_1 b_3) \hat{j} + (c_2 b_1 - c_1 b_2) \hat{k}$$

$$a \cdot (b \times c) = a_1 b_2 c_3 - a_1 c_2 b_3 - b_1 a_2 c_3 + c_1 a_2 b_3 + b_1 c_2 a_3 - c_1 b_2 a_3$$

RHS:

$(a \times b) \cdot c$

$$a \times b = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= (a_2 b_3 - b_2 a_3) \hat{i} - (a_1 b_3 - b_1 a_3) \hat{j} + (a_1 b_2 - b_1 a_2) \hat{k}$$

$$(a \times b) \cdot c = c_1 a_2 b_3 - c_1 b_2 a_3 - a_1 c_2 b_3 + b_1 c_2 a_3 + a_1 b_2 c_3 - b_1 a_2 c_3$$

LHS = RHS (Hence Proved).

b)  $a \times (b \times c) = (a \cdot c) b - (a \cdot b) c$

LHS:

$a \times (b \times c)$

$$b \times c = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = (c_3 b_2 - c_2 b_3) \hat{i} - (b_1 c_3 - c_1 b_3) \hat{j} + (c_2 b_1 - c_1 b_2) \hat{k}$$

$$a \times (b \times c) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ c_3 b_2 - c_2 b_3 & b_1 c_3 - c_1 b_3 & c_2 b_1 - c_1 b_2 \end{vmatrix}$$

$$= (-a_2 c_2 b_1 + c_1 a_2 b_2 + b_1 a_3 c_3 - c_1 a_3 b_3) \hat{i} - (a_1 b_1 c_2 + a_1 c_1 b_2 + b_2 a_3 c_3 - c_2 b_3 a_3) \hat{j} + (-a_1 b_1 c_3 + a_1 c_1 b_3 + a_2 b_2 c_3 - a_2 c_2 b_3) \hat{k}$$

RHS:

(a · c) · b

$$(a \cdot c) \cdot b - (a \cdot b) \cdot c$$

$$a \cdot c = a_1 c_1 + a_2 c_2 + a_3 c_3$$

$$\begin{aligned}(a \cdot c) \cdot b &= (a_1 c_1 + a_2 c_2 + a_3 c_3) \cdot [b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}] \\&= (a_1 b_1 c_1 + b_1 a_2 c_2 + b_1 a_3 c_3) \hat{i} + (a_1 c_1 b_2 + a_2 b_2 c_2 + b_2 a_3 c_3) \hat{j} \\&\quad + (a_1 c_1 b_3 + a_2 b_2 b_3 + a_3 b_3 c_3) \hat{k}\end{aligned}$$

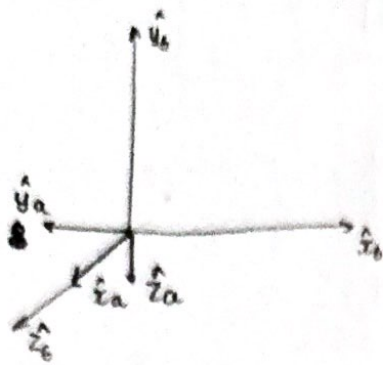
$$a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\begin{aligned}(a \cdot b) \cdot c &= (a_1 b_1 + a_2 b_2 + a_3 b_3) [c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}] \\&= (a_1 b_1 c_1 + c_1 a_2 b_2 + c_1 a_3 b_3) \hat{i} + (a_1 b_1 c_2 + a_2 b_2 c_2 + c_2 a_3 b_3) \hat{j} \\&\quad + (a_1 b_1 c_3 + a_2 b_2 c_3 + a_3 b_3 c_3) \hat{k}\end{aligned}$$

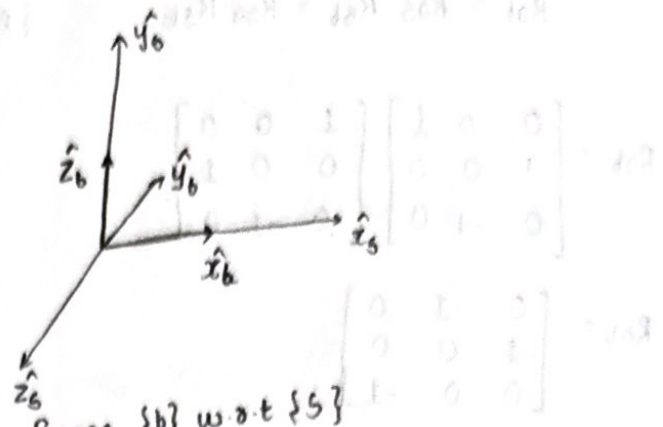
$$\begin{aligned}\therefore (a \cdot c) \cdot b - (a \cdot b) \cdot c &= [a_1 b_1 c_1 + b_1 a_2 c_2 + b_1 a_3 c_3 - \cancel{a_1 b_1 c_1} - c_1 a_2 b_2 - c_1 a_3 b_3] \hat{i} \\&\quad + [a_1 c_1 b_2 + \cancel{a_2 b_2 c_2} + b_2 a_3 c_3 - \cancel{a_1 b_1 c_2} - \cancel{a_2 b_2 c_2} - c_2 a_3 b_3] \hat{j} \\&\quad + [a_1 c_1 b_3 + \cancel{a_2 b_2 b_3} + \cancel{a_3 b_3 c_3} - \cancel{a_1 b_1 c_3} - \cancel{a_2 b_2 c_3} - \cancel{a_3 b_3 c_3}] \hat{k} \\&= [b_1 a_2 c_2 + b_1 a_3 c_3 - c_1 a_2 b_2 - c_1 a_3 b_3] \hat{i} + [a_1 c_1 b_2 + b_2 a_3 c_3 - a_1 b_1 c_2 - c_2 a_3 b_3] \hat{j} \\&\quad + [a_1 c_1 b_3 + a_2 b_2 b_3 - a_1 b_1 c_3 - a_2 b_2 c_3] \hat{k}\end{aligned}$$

$\therefore \text{LHS} = \text{RHS}$  [Hence Proved]





- Frame {a} w.r.t to {S}



- Frame {b} w.r.t {S}

b)

$$R_{Sa} = \begin{bmatrix} x_a \cdot x_s & y_a \cdot x_s & z_a \cdot x_s \\ x_a \cdot y_s & y_a \cdot y_s & z_a \cdot y_s \\ x_a \cdot z_s & y_a \cdot z_s & z_a \cdot z_s \end{bmatrix}$$

$$R_{Sa} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$R_{Sb} = \begin{bmatrix} x_b \cdot x_s & y_b \cdot x_s & z_b \cdot x_s \\ x_b \cdot y_s & y_b \cdot y_s & z_b \cdot y_s \\ x_b \cdot z_s & y_b \cdot z_s & z_b \cdot z_s \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

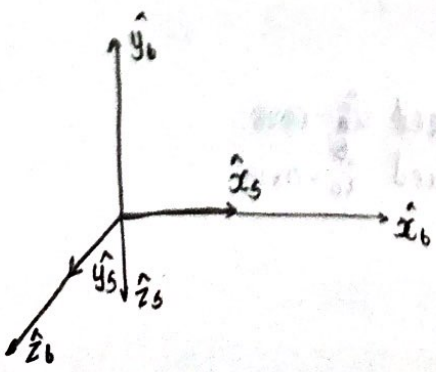
$$R = R_{Sp} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

c) For rotation matrix ( $R \in SO(3)$ )  $R^T = R^{-1}$

$$\therefore R_{Sb}^{-1} = R_{Sb}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



d)

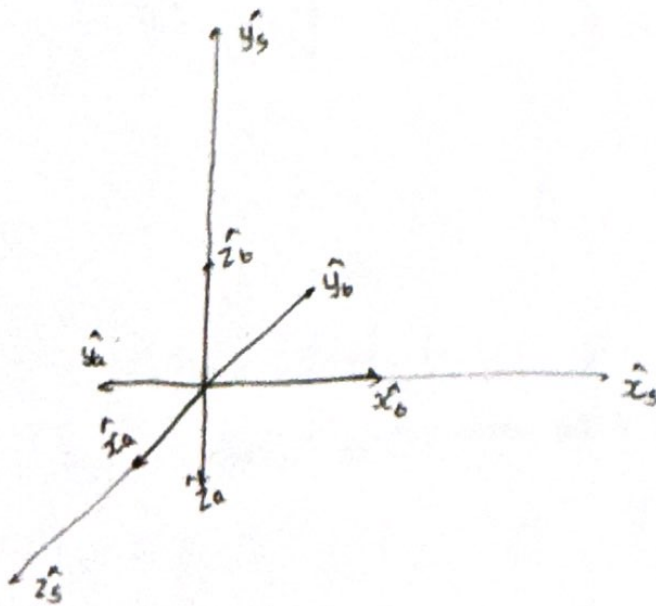
from matrix rotation property

$$R_{ab} = R_{as} \cdot R_{sb} = R_{sa}^T R_{sb}$$

$$\{R_{as} = R_{sa}^T\}$$

$$R_{ab} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$R_{ab} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$



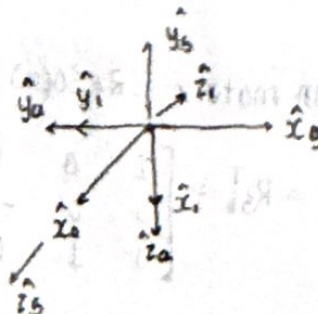
e)

$$R = R_{sb} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$R_1 = R_{sa} \cdot R = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

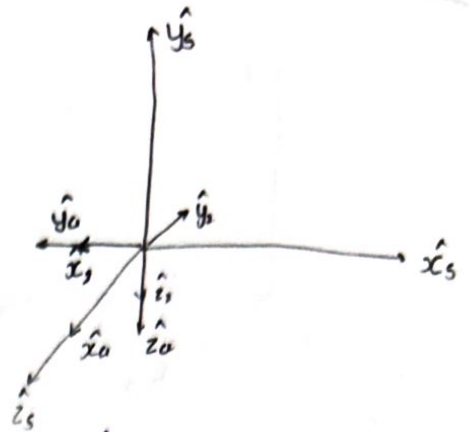
$$= \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

→  $R_1$  correspond to rotation of  $R_{sa}$  by  $-90^\circ$  about   
 body-fixed  $\hat{y}_b$ -axis.   
 body-fixed  $\hat{x}_b$ -axis



$$R_2 = R_{Sb} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



-  $R_2$  correspond of  $R_{Sb}$  by  $-90^\circ$  about  $\hat{x}_b$ -axis would be fixed.

f)

$$P_5 = R_{Sb} P_b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

g)

To rep  $P_b$  in  $\{S\}$

$$P^I = R_{Sb} P_b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

•  $P^I$  represent moving location of point withing same reference frame  $\{S\}$ .

$$P^{II} = R_{Sb}^T P_b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

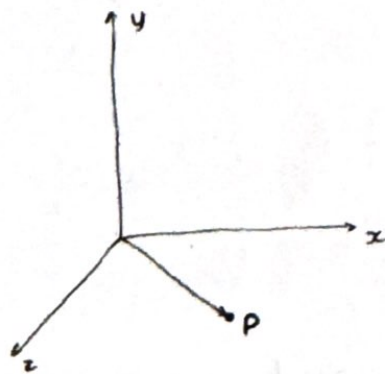
•  $P^{II}$  (which is  $R_{bS} P_b$ ) rep changing co-ordinates to frame  $\{b\}$  without moving physical point.

h)

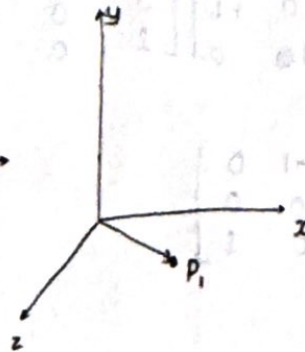
$w$  is a vector that can be rotated similarly as a position vector

$$w_2 = R_{Sb}^T w_b = R_{Sb} w_b = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix}$$

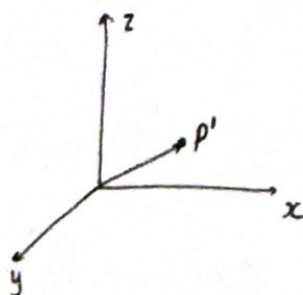




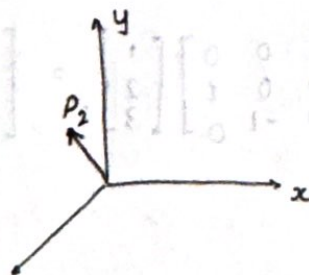
$30^\circ$   
about x



$135^\circ$   
about y



$-120^\circ$   
about z



(a)

$$P = \left[ \frac{1}{\sqrt{3}} \quad \frac{1}{\sqrt{6}} \quad \frac{1}{\sqrt{2}} \right]^T$$

i) Rotation about x-axis by  $30^\circ$

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 30^\circ & -\sin 30^\circ \\ 0 & \sin 30^\circ & \cos 30^\circ \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

ii) Rotation about y-axis by  $135^\circ$

$$R_y = \begin{bmatrix} \cos 135^\circ & 0 & \sin 135^\circ \\ 0 & 1 & 0 \\ -\sin 135^\circ & 0 & \cos 135^\circ \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

iii) Rotation about z-axis by  $-120^\circ$

$$R_z = \begin{bmatrix} \cos 120^\circ & -\sin 120^\circ & 0 \\ \sin 120^\circ & \cos 120^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore P_1 = R_x P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$$

$$\therefore P_2 = R_y P_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}-\sqrt{6}}{6} \\ -\frac{1}{\sqrt{2}} \\ -\frac{\sqrt{3}-\sqrt{6}}{6} \end{bmatrix}$$

$$\therefore P' = R_z P_2 = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}-\sqrt{6}}{6} \\ -\frac{1}{\sqrt{2}} \\ -\frac{\sqrt{3}-\sqrt{6}}{6} \end{bmatrix} = \begin{bmatrix} \frac{-\sqrt{3}-2\sqrt{6}}{12} \\ \frac{-1+2\sqrt{2}}{4} \\ \frac{-\sqrt{3}-\sqrt{6}}{6} \end{bmatrix}$$

$$\therefore P' = \begin{bmatrix} \frac{-\sqrt{3}-2\sqrt{6}}{12} & \frac{-1+2\sqrt{2}}{4} & \frac{-\sqrt{3}-\sqrt{6}}{6} \end{bmatrix}^T$$

(b)

$$R = R_x R_y R_z$$

$$R = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$R = \begin{bmatrix} \frac{1}{2\sqrt{2}} & \frac{\sqrt{3}}{2} & \frac{-1}{2\sqrt{2}} \\ \frac{\sqrt{3}}{2\sqrt{2}} & -\frac{1}{2} & \frac{-\sqrt{3}}{2\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$R = \begin{bmatrix} \frac{1}{2\sqrt{2}} & \frac{6-\sqrt{2}}{8} & \frac{-2\sqrt{3}+\sqrt{6}}{8} \\ \frac{\sqrt{3}}{2\sqrt{2}} & \frac{-2\sqrt{3}+\sqrt{6}}{8} & \frac{2-3\sqrt{2}}{8} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{2\sqrt{2}} & -\frac{\sqrt{3}}{2\sqrt{2}} \end{bmatrix}$$



P-4.

Given

- 3 frames:  $\{\hat{x}_a \hat{y}_a \hat{z}_a\}$ ;  $\{\hat{x}_b \hat{y}_b \hat{z}_b\}$ ;  $\{\hat{x}_c \hat{y}_c \hat{z}_c\}$ .

$$\begin{aligned}\hat{x}_b &= \delta_{11} \hat{x}_a + \delta_{21} \hat{y}_a + \delta_{31} \hat{z}_a \\ \hat{y}_b &= \delta_{12} \hat{x}_a + \delta_{22} \hat{y}_a + \delta_{32} \hat{z}_a \\ \hat{z}_b &= \delta_{13} \hat{x}_a + \delta_{23} \hat{y}_a + \delta_{33} \hat{z}_a\end{aligned}$$

$$\begin{aligned}\hat{x}_c &= \delta_{11} \hat{x}_b + \delta_{21} \hat{y}_b + \delta_{31} \hat{z}_b \\ \hat{y}_c &= \delta_{12} \hat{x}_b + \delta_{22} \hat{y}_b + \delta_{32} \hat{z}_b \\ \hat{z}_c &= \delta_{13} \hat{x}_b + \delta_{23} \hat{y}_b + \delta_{33} \hat{z}_b\end{aligned}$$

$\therefore$  From given eq<sup>n</sup> of  $\{\hat{x}_b \hat{y}_b \hat{z}_b\}$  in  $\{a\}$ , matrix  $R_a^b$  is

$$R_a^b = \begin{bmatrix} \delta_{11} & \delta_{12} & \delta_{13} \\ \delta_{21} & \delta_{22} & \delta_{23} \\ \delta_{31} & \delta_{32} & \delta_{33} \end{bmatrix}$$

From given eq<sup>n</sup> of  $\{\hat{x}_c \hat{y}_c \hat{z}_c\}$  in  $\{b\}$

$$R_b^c = \begin{bmatrix} \delta_{11} & \delta_{12} & \delta_{13} \\ \delta_{21} & \delta_{22} & \delta_{23} \\ \delta_{31} & \delta_{32} & \delta_{33} \end{bmatrix}$$

To verify  $R_a^b \cdot R_b^c = R_a^c$ , we will need to write  $\{\hat{x}_c \hat{y}_c \hat{z}_c\}$  in  $\{a\}$ .

$$R_a^c = \begin{bmatrix} \delta_{11} & \delta_{12} & \delta_{13} \\ \delta_{21} & \delta_{22} & \delta_{23} \\ \delta_{31} & \delta_{32} & \delta_{33} \end{bmatrix} \begin{bmatrix} \delta_{11} & \delta_{12} & \delta_{13} \\ \delta_{21} & \delta_{22} & \delta_{23} \\ \delta_{31} & \delta_{32} & \delta_{33} \end{bmatrix}$$

$$R_a^c = \begin{bmatrix} \delta_{11}\delta_{11} + \delta_{12}\delta_{21} + \delta_{13}\delta_{31} & \delta_{11}\delta_{12} + \delta_{12}\delta_{22} + \delta_{13}\delta_{32} & \delta_{11}\delta_{13} + \delta_{12}\delta_{23} + \delta_{13}\delta_{33} \\ \delta_{21}\delta_{11} + \delta_{22}\delta_{21} + \delta_{23}\delta_{31} & \delta_{21}\delta_{12} + \delta_{22}\delta_{22} + \delta_{23}\delta_{32} & \delta_{21}\delta_{13} + \delta_{22}\delta_{23} + \delta_{23}\delta_{33} \\ \delta_{31}\delta_{11} + \delta_{32}\delta_{21} + \delta_{33}\delta_{31} & \delta_{31}\delta_{12} + \delta_{32}\delta_{22} + \delta_{33}\delta_{32} & \delta_{31}\delta_{13} + \delta_{32}\delta_{23} + \delta_{33}\delta_{33} \end{bmatrix}$$



$$\hat{x}_c = s_{11} \hat{x}_b + s_{21} \hat{y}_b + s_{31} \hat{z}_b$$

Substitute  $\hat{x}_b, \hat{y}_b, \hat{z}_b$

$$\hat{x}_c = s_{11}(\hat{x}_a + \delta_{11} \hat{x}_a + \delta_{21} \hat{y}_a + \delta_{31} \hat{z}_a)$$

$$\hat{x}_c = s_{11}(\delta_{11} \hat{x}_a + \delta_{21} \hat{y}_a + \delta_{31} \hat{z}_a) + s_{21}(\delta_{12} \hat{x}_a + \delta_{22} \hat{y}_a + \delta_{32} \hat{z}_a) + s_{31}(\delta_{31} \hat{x}_a + \delta_{32} \hat{y}_a + \delta_{33} \hat{z}_a)$$

$$\hat{x}_c = (s_{11}\delta_{11} + s_{21}\delta_{12} + s_{31}\delta_{31})\hat{x}_a + (s_{11}\delta_{21} + s_{21}\delta_{22} + s_{31}\delta_{32})\hat{y}_a + (s_{11}\delta_{31} + s_{21}\delta_{32} + s_{31}\delta_{33})\hat{z}_a$$

→ Similar with:  $\hat{y}_c = s_{12} \hat{x}_b + s_{22} \hat{y}_b + s_{32} \hat{z}_b$

$$\hat{y}_c = s_{12}(\delta_{11} \hat{x}_a + \delta_{21} \hat{y}_a + \delta_{31} \hat{z}_a) + s_{22}(\delta_{12} \hat{x}_a + \delta_{22} \hat{y}_a + \delta_{32} \hat{z}_a) + s_{32}(\delta_{31} \hat{x}_a + \delta_{32} \hat{y}_a + \delta_{33} \hat{z}_a)$$

$$\hat{y}_c = (s_{12}\delta_{11} + s_{22}\delta_{12} + s_{32}\delta_{31})\hat{x}_a + (s_{12}\delta_{21} + s_{22}\delta_{22} + s_{32}\delta_{32})\hat{y}_a + (s_{12}\delta_{31} + s_{22}\delta_{32} + s_{32}\delta_{33})\hat{z}_a$$

→ Similar with:  $\hat{z}_c = s_{13} \hat{x}_b + s_{23} \hat{y}_b + s_{33} \hat{z}_b$

$$\hat{z}_c = s_{13}(\delta_{11} \hat{x}_a + \delta_{21} \hat{y}_a + \delta_{31} \hat{z}_a) + s_{23}(\delta_{12} \hat{x}_a + \delta_{22} \hat{y}_a + \delta_{32} \hat{z}_a) + s_{33}(\delta_{31} \hat{x}_a + \delta_{32} \hat{y}_a + \delta_{33} \hat{z}_a)$$

$$\hat{z}_c = (s_{13}\delta_{11} + s_{23}\delta_{12} + s_{33}\delta_{31})\hat{x}_a + (s_{13}\delta_{21} + s_{23}\delta_{22} + s_{33}\delta_{32})\hat{y}_a + (s_{13}\delta_{31} + s_{23}\delta_{32} + s_{33}\delta_{33})\hat{z}_a$$

∴ Rotation matrix formed by  $\{\hat{x}_c, \hat{y}_c, \hat{z}_c\}$  in  $\{a\}$  frame:

$$R = \begin{bmatrix} s_{11}\delta_{11} + s_{21}\delta_{12} + s_{31}\delta_{31} & s_{11}\delta_{21} + s_{21}\delta_{22} + s_{31}\delta_{32} & s_{11}\delta_{31} + s_{21}\delta_{32} + s_{31}\delta_{33} \\ s_{12}\delta_{11} + s_{22}\delta_{12} + s_{32}\delta_{31} & s_{12}\delta_{21} + s_{22}\delta_{22} + s_{32}\delta_{32} & s_{12}\delta_{31} + s_{22}\delta_{32} + s_{32}\delta_{33} \\ s_{13}\delta_{11} + s_{23}\delta_{12} + s_{33}\delta_{31} & s_{13}\delta_{21} + s_{23}\delta_{22} + s_{33}\delta_{32} & s_{13}\delta_{31} + s_{23}\delta_{32} + s_{33}\delta_{33} \end{bmatrix}$$

∴ we can see:  $R = R^a$

Hence:  $R^a \cdot R^b = R^c$

Hence proved.

P-5

(a)

To prove:  $R(v \times w) = (Rv) \times (Rw)$

→ Where  $R \in SO(3)$  &  $v, w \in \mathbb{R}^3$

∴ As  $R \in SO(3)$

$RR^T = I$  and  $\det(R) = 1$ .

→ Let  $u$  be an vector  $u \in \mathbb{R}^3$

Taking RHS & doing it with  $u$ .

$$u \cdot ((Rv) \times (Rw))$$

• Taking determinant Geometrically dot product with scalar vectors cross product form a scalar triple product, which is equivalent to  $3 \times 3$  matrix formed by placing vectors as column.

$$\det[u \ Rv \ Rw]$$

$$\therefore \det[u \ Rv \ Rw] = \det[Iu \ Rv \ Rw]$$

$$= \det[RR^T u \ Rv \ Rw]$$

$$= \det[R(R^T u \ v \ w)]$$

$$\therefore (\det(A) \cdot \det(B)) = \det(AB)$$

$$\therefore \det[R(R^T u \ v \ w)] = \det(R) \cdot \det[R^T u \ v \ w]$$

$$\therefore (\det(R) = 1)$$

$$\therefore \det(R) \cdot \det[R^T u \ v \ w] = \det[R^T u \ v \ w]$$

• The determinant  $\det[R^T u \ v \ w]$  can be written in dot product using same argument given above.

$$\therefore \det[R^T u \ v \ w] = (R^T u) \cdot (v \times w)$$

$$(R^T u) \cdot (v \times w) = u \cdot (R(v \times w))$$

$$\therefore (Ax) \cdot y = x \cdot (A^T y)$$

$$\therefore u \cdot (R^T(v \times w)) = u \cdot (R(v \times w))$$

∴ we have shown  $u \cdot ((Rv) \times (Rw)) = u \cdot (R(v \times w))$  for any vector  $u$ .



$\therefore$  We have shown:  $u((Rv) \times (Rw)) = u(R(v \times w))$  for absolutely any vector  $u$ , and for this to be true, the vector component itself must be identical.  $\Rightarrow R(v \times w) = (Rv) \times (Rw)$ .

b)

for  $w \in \mathbb{R}^3$

Prove that:  $R \hat{w} R^T = (R\hat{w})$

Taking LHS and multiplying it with any vector  $v$ ;  $v \in \mathbb{R}^3$

$$(R \hat{w} R^T) v = R(\hat{w} (R^T v))$$

$$\therefore (\hat{a} b = a \times b)$$

$$R(\hat{w} (R^T v)) = R(\hat{w} \times (R^T v))$$

$\therefore$  from previous part (a) derivation:  $R(a \times b) = (Ra) \times (Rb)$

$$\therefore R(\hat{w} \times (R^T v)) = (Rw) \times (R(R^T v))$$

$$R(\hat{w} \times (R^T v)) = Rw \times (Iv)$$

$$R(\hat{w} \times (R^T v)) = Rw \times v$$

$$R(\hat{w} \times (R^T v)) = (R\hat{w}) v$$

$$\therefore (R \hat{w} R^T) v = (R\hat{w}) v \quad ; \text{ true for any } v \in \mathbb{R}^3$$

$\rightarrow$  As this transformation hold true for any vector  $v$  the transforming matrices must be same:

$$\therefore R \hat{w} R^T = R\hat{w}$$

c)  $R = [\hat{r}_1 \ \hat{r}_2 \ \hat{r}_3] \quad ; \ R \in SO(3)$

Prove that:  $\det(R) = \hat{r}_1^T (\hat{r}_2 \times \hat{r}_3)$ .

- Here we can use properties of rotation matrix.

①  $\det(R) = 1$

② column of rotation matrix are orthonormal, linearly independent & each independent column vector is a unit vector.

Let:  $\gamma_1 = [a_1 \ a_2 \ a_3]^T$

$\gamma_2 = [b_1 \ b_2 \ b_3]^T$

$\gamma_3 = [c_1 \ c_2 \ c_3]^T$

$$\therefore (\gamma_2 \times \gamma_3) = \begin{bmatrix} b_2 c_3 - b_3 c_2 \\ b_3 c_1 - b_1 c_3 \\ b_1 c_2 - b_2 c_1 \end{bmatrix}$$

$\Rightarrow$  now as  $\gamma_2$  &  $\gamma_3$  are orthonormal their cross product would be equal to  $\gamma_1$ . {Using Right hand rule}

$$\Rightarrow \begin{bmatrix} b_2 c_3 - b_3 c_2 \\ b_3 c_1 - b_1 c_3 \\ b_1 c_2 - b_2 c_1 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$\therefore \gamma_1^T \cdot \gamma_1 = [a_1 \ a_2 \ a_3] \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$\gamma_1^T \cdot \gamma_1 = [a_1^2 + a_2^2 + a_3^2]$$

$\therefore$  As  $\gamma_1$  is unit vector:  $a_1^2 + a_2^2 + a_3^2 = 1$ .

$\therefore (\gamma_1^T \cdot \gamma_1) = 1 = \det(R)$ .

$\therefore$  Hence proved.

(d)

Given is  $so(3)$  set of skew-symmetric matrix

$\therefore so(3) = [A \in M_{3 \times 3}(\mathbb{R}) \mid A^T = -A]$  ;  $M_{3 \times 3}$  is a  $3 \times 3$  vector space.

$\rightarrow$  As  $so(3)$  is subset of standard vector space  $3 \times 3$  real matrices  $M_{3 \times 3}(\mathbb{R})$  we can show it as a subspace by verifying closure under addition & scalar multiplication.



• Non-empty (zero matrix):

$3 \times 3$  zero matrix  $O$  satisfy:  $O^T = O = -O \Rightarrow O \in \mathfrak{so}(3)$ .

• Addition closure:

$$A, B \in \mathfrak{so}(3) \Rightarrow A^T = -A \text{ \& } B^T = -B$$

$$(A+B)^T = A^T + B^T$$

$$= -A - B$$

$$= -(A+B)$$

$$\Rightarrow (A+B) \in \mathfrak{so}(3)$$

• scalar multiplication closure:

$$A \in \mathfrak{so}(3) \Rightarrow A^T = -A$$

$$(cA)^T = c(A^T)$$

$$= c(-A)$$

$$= -(cA)$$

$$\Rightarrow (cA) \in \mathfrak{so}(3)$$

$\therefore$  Since  $\mathfrak{so}(3)$  contain zero matrix & it's closure under addition & scalar multiplication, it is vector space.

$\Rightarrow$  Now for dimension & basis:-

$$\text{If } A \in \mathfrak{so}(3) \Rightarrow A^T = -A$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} ; -A = \begin{bmatrix} -a_{11} & -a_{12} & -a_{13} \\ -a_{21} & -a_{22} & -a_{23} \\ -a_{31} & -a_{32} & -a_{33} \end{bmatrix}$$

$$\text{As } A^T = -A \Rightarrow a_{11} = -a_{11}, a_{22} = -a_{22}, a_{33} = -a_{33}$$

this is only possible when  $a_{11} = a_{22} = a_{33} = 0$ .

& off diagonal elements:-

$$a_{21} = -a_{12} ; a_{31} = -a_{13} ; a_{32} = -a_{23}$$

$\therefore$  A matrix can be made of using 3 real no  $x_1, x_2, x_3$ .

$$\therefore A = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}$$

$$\therefore A = x_1 \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}}_{E_1} + x_2 \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}}_{E_2} + x_3 \underbrace{\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{E_3}$$

$\therefore$  Standard basis for  $so(3)$  is given by set  $B = \{E_1, E_2, E_3\}$

Since  $E_1, E_2, E_3$  ~~span~~ linearly span  $so(3)$

$$x_1 E_1 + x_2 E_2 + x_3 E_3 = 0 \Rightarrow x_1 = x_2 = x_3 = 0$$

$$\therefore \underline{\dim(so(3)) = 3.}$$

P-6

(a) Let  $R = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in so(2)$

$$\Rightarrow R^T R = I \text{ and } \det(R) = 1.$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + b^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\therefore a^2 + b^2 = 1$ , the first column vector  $(a, c)$  lie on unit circle  $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ .

$\therefore$  there must be an angle  $\theta \in [0, 2\pi)$

$$a = \cos \theta, c = \sin \theta.$$

$$\therefore ab + cd = 0 \text{ \& } \det(R) = ad - bc = 1.$$

$$b \cos \theta + d \sin \theta = 0 \quad \dots \textcircled{1}$$

$$d \cos \theta - b \sin \theta = 1 \quad \dots \textcircled{2}$$



→ multiply ② with 'd' & ① with b.

$$\begin{array}{r} b^2 \cos \theta + b d \sin \theta = 0 \\ d^2 \cos \theta - b d \sin \theta = d \\ \hline \cos \theta (b^2 + d^2) + 0 = d. \end{array}$$

$$\Rightarrow d = \cos \theta$$

$$\therefore b = -\sin \theta.$$

$$\therefore R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

The mapping of  $f: \text{SO}(2) \rightarrow S^1$  defined by  $f(R(\theta)) = (\cos \theta, \sin \theta)$  is a bijection.  
Hence  $\text{SO}(2)$  is identified as  $S^1$ .

(b) for  $R \in \text{SO}(2)$  &  $\hat{w} \in \text{SO}(2)$

$$\text{P.T: } R \hat{w} R^T = \hat{w}$$

$\text{SO}(2)$  contain skew symmetric matrix.

$\therefore$  from above derivation.

$$\hat{w} = \begin{bmatrix} 0 & -w \\ w & 0 \end{bmatrix}; w \in \mathbb{R}.$$

$\therefore$  Pre multiply  $R$  with  $\hat{w}$ .

$$\begin{aligned} R \hat{w} &= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 0 & -w \\ w & 0 \end{bmatrix} \\ &= \begin{bmatrix} -w \sin \theta & -w \cos \theta \\ w \cos \theta & -w \sin \theta \end{bmatrix} \end{aligned}$$

$\therefore$  Post multiply  $R \hat{w}$  with  $R^T$ .

$$R \hat{w} R^T = \begin{bmatrix} -w \sin \theta & -w \cos \theta \\ w \cos \theta & -w \sin \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$R \hat{w} R^T = \begin{bmatrix} -w \cos \theta \sin \theta + w \cos \theta \sin \theta & -w \sin^2 \theta - w \cos^2 \theta \\ w \cos^2 \theta + w \sin^2 \theta & w \cos \theta \sin \theta - w \cos \theta \sin \theta \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -w \\ w & 0 \end{bmatrix}$$

$$R \hat{\omega} R^T = \begin{bmatrix} 0 & -\omega(\sin^2\theta + \cos^2\theta) \\ \omega(\sin^2\theta + \cos^2\theta) & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} = \hat{\omega}$$

$\therefore R \hat{\omega} R^T = \hat{\omega}$  ; Hence Proved.



**Tool used:**

Google Gemini

**What I asked:**

- Problem 2: This was mostly formula-based (finding rotation matrices from given axis directions), so I solved the entire problem by hand first, then asked Gemini to solve it independently just to confirm my final matrices and answers matched.
- Problem 3: This was also a purely formula-based approach question, so I solved the entire problem by hand first, then asked Gemini to solve it independently just to confirm my final matrices and answers matched.
- Problem 4: Asked a general prompt on how to prove  $\mathbf{R}_{ab} \cdot \mathbf{R}_{bc} = \mathbf{R}_{ac}$  given the unit vectors of one frame expressed in terms of another. Separately asked it to solve the problem to compare against my own term-by-term substitution proof.
- Problem 5: Asked general prompts on properties of rotation matrices and vector identities — how the scalar triple product relates to determinants, how orthonormality of rotation matrix columns can be used in proofs, and how the skew-symmetric (hat) operator relates to the cross product. Separately asked it to solve parts (a)–(d) fully to check my proofs.
- Problem 6: Asked a general prompt on how  $SO(2)$  matrices relate to points on the unit circle, and how the orthogonality/determinant conditions constrain a  $2 \times 2$  rotation matrix. Separately asked it to solve parts (a) and (b) fully to verify against my derivation.

**How I used the output (and what I changed/verified myself):**

- Used the general property/concept prompts only as a refresher on definitions and identities (triple product, orthonormality, skew-symmetric operator), not to get an approach for the specific problem.
- Worked out my own approach and full derivation for each problem independently, before or after these general prompts.
- Compared Gemini's fully solved versions against my own hand-worked solutions, only to confirm final answers and matrices matched.
- Did not copy derivation steps or wording. My proofs use my own notation and, in some cases (e.g., Problem 5(c), which uses orthonormal columns), a different, valid method than the one Gemini used.
- Kept my original hand-drawn sketches for Problem 2 rather than redrawing based on Gemini's description.